

# Logic hazards

Combinational logic can glitch when inputs change

# One input moves, paths differ

- Hazard analysis usually changes one input at a time
- Different path delays let one SOP term drop before another picks up
- A consensus term that bridges adjacent cubes, like  $BC$  here, can cover the gap
- Equal delays can hide the glitch in simulation but not in silicon

# Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

## Static / dynamic hazards

Single-input transitions on multi-level logic can glitch. Spot **static-1**, **static-0**, and **dynamic** hazards — and cover them.

**Starter example:**  $F = AB + A'C$ , flip A:  $1 \rightarrow 0$  with  $B=C=1$ . Term AB falls before A'C rises  $\rightarrow$  **static-1** glitch. Cover with consensus + BC.

Load starter example

### Challenges (1/22)

Quiz: **static-1**. Stay-1 with a dip to 0 is a? Answer: static -1

Show hintCheckNextIdle

quiz-static1

quiz-static0

quiz-dynamic

quiz-consensus

starter-load

see-static1

cover-apply

cover-safe

quiz-term

static0-preset

see-static0

dynamic-preset

see-dynamic

quiz-single

explain

delay-gap

equal-delay

quiz-sop

quiz-pos

flip-hold

run-btn

full-fix

### Core ideas

|                                                          |                                                          |                                                                 |
|----------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------|
| <b>Static-1</b><br>Output should stay 1, briefly goes 0. | <b>Static-0</b><br>Output should stay 0, briefly goes 1. | <b>Dynamic</b><br>More than one edge when the function has one. |
|----------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------|

Logic hazards starter

# Workbook practice

- In the workbook track
- Write the consensus term  $BC$  by hand
- Name static-one versus static-zero from the dip or spike direction
- Note one pitfall: assuming equal gate delays means no hazard in real hardware

# Pitfalls to watch

- Do not confuse a hazard with wrong logic
- Cover terms must actually bridge the risky transition
- And remember: the browser lab is literacy
- Real designs still need timing analysis

# Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, identify one static hazard and one consensus fix
- When you are ready, take the short quiz, then continue to gate composer